Lab 08

Thevenin's, Norton's and Maximum Power Transfer Theorems

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8.1 Objectives:

The Objectives of this Lab are:

- To verify the Thevenin, Norton and Maximum Power Transfer theorems when applied AC circuits.

8.2 Pre-Lab

Task 1

1. Calculate V_{LOAD}, V_{TH} and Z_{TH} of circuit given in Figure 8.1 and fill the spaces in Task 2 and Table 8.1.
2. Calculate I_{LOAD}, I_N and Z_N of circuit given in Figure 8.2 and fill Table 8.2.
3. Calculate the Z_{TH} and Z_{LOAD}[1] of circuit given in Figure 8.3.

Introduction

The aim of this Lab is to apply the Electric Circuits Theorems. Electric circuit theorems are always beneficial to help find voltage and currents in multi-loop circuits. These theorems use fundamental rules or formulas and basic equations of mathematics to analyze basic components of electrical or electronics parameters such as voltages, currents, resistance, and so on. These fundamental theorems include the basic theorems like Superposition theorem, Tellegen's theorem, Norton's theorem, Maximum power transfer theorem, and Thevenin's theorems. In this Lab you are going to learn the application of three of them which includes, Thevenin, Norton and Maximum Power Transfer theorems when AC voltage is applied.

8.3 Thevenin's Theorem

Thevenin's theorem states that any linear two port network can be replaced by a single voltage source with series impedance. While the theorem is applicable to any number of voltage and current sources, this exercise will only examine single source circuits for simplicity of the circuit. The Thevenin voltage is the open circuit output voltage. This may be determined experimentally by isolating the portion to be thevenized and simply placing an oscilloscope at its output terminals. The Thevenin impedance is found by replacing all sources with their internal impedance and then applying appropriate series-parallel impedance simplification rules.

[1] Read the manual section 8.4

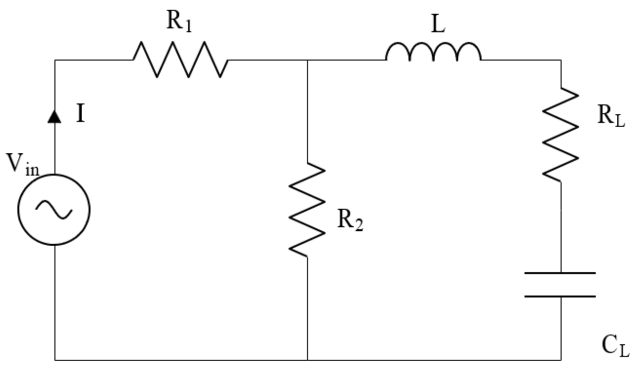


Figure 8.1: Schematic for AC circuit

Task 2: To simulate the Thevenin's voltage and impedance of the given AC Circuit.

Note: Attach snippet of every circuit and graph, all the graphs and circuits must be labelled properly.

1. For the circuit of Figure 8.1 record the voltage across the load $(\boldsymbol{R_L = 1\ k\Omega\ and\ C_L = 47nF})$. Use $R1 = 1.5\ k\Omega,\ R2 = 2.2\ k\Omega$ and $L = 10mH$ with an AC voltage source of value $1V$ and frequency $10kHz$. Record the voltage in given space below:

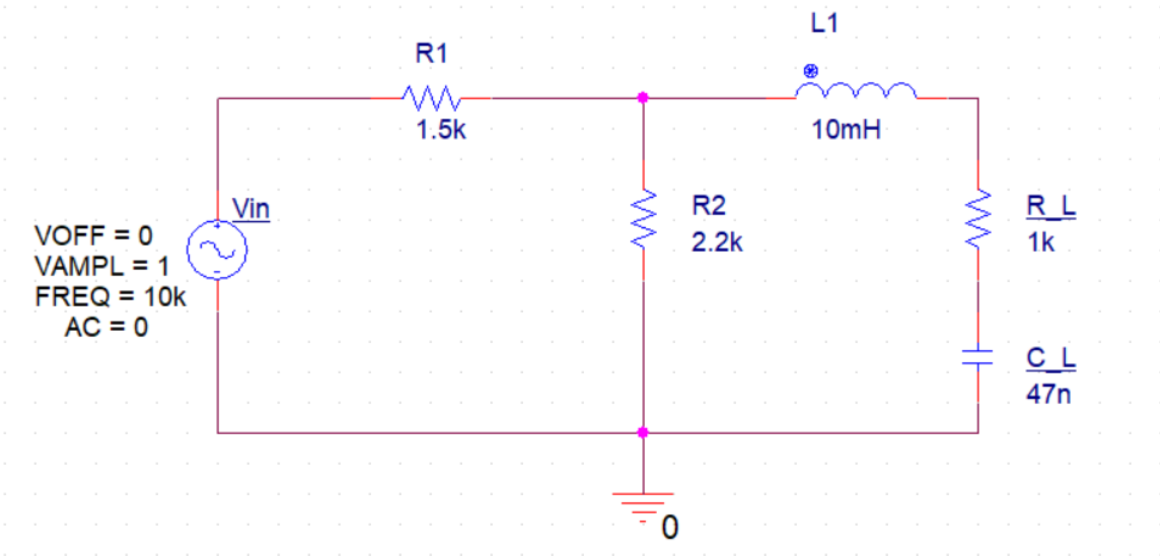
   V_{LOAD} (Calculated) = 0.328 < -27.4

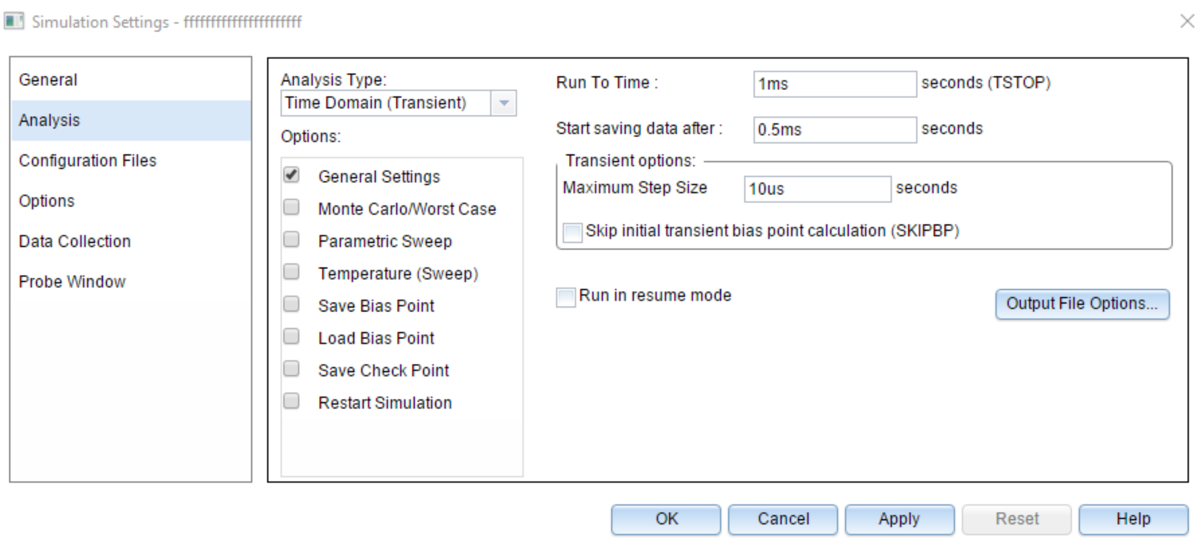
2. Record the expected Thevenin's voltage (V_{TH}) and Thevenin's impedance (Z_{TH}).

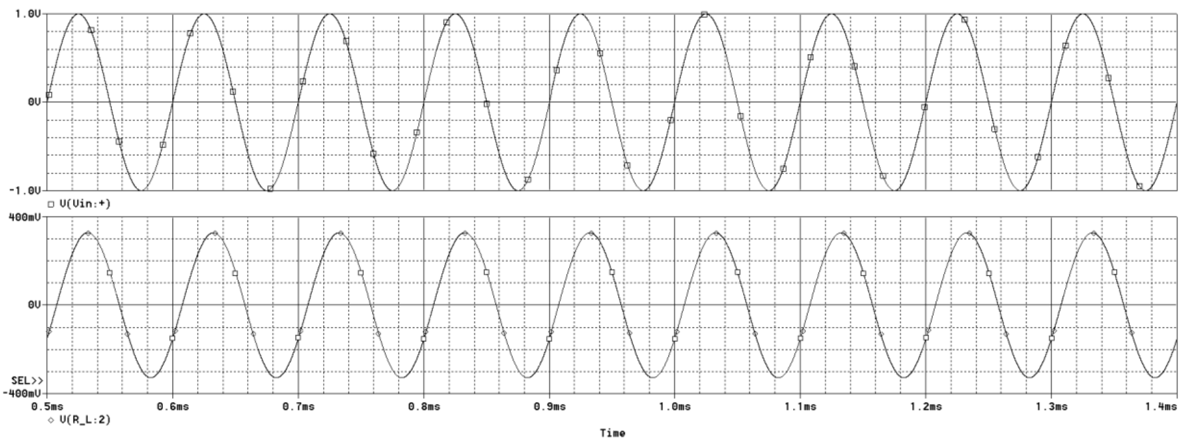
   V_{TH} (Calculated) = 0.5946V

   Z_{TH} (Calculated) = 1090.9<35.1

3. Simulate the circuit given in Figure 8.1 Measure the load voltage (V_L) and Δt (w.r.t V_{IN}) in given space below. Make sure you measure Δt and ϕ with correct sign. Discard first 5 cycles then take measurement. For smother waveform keep the maximum step size in 100 times smaller than the time period.



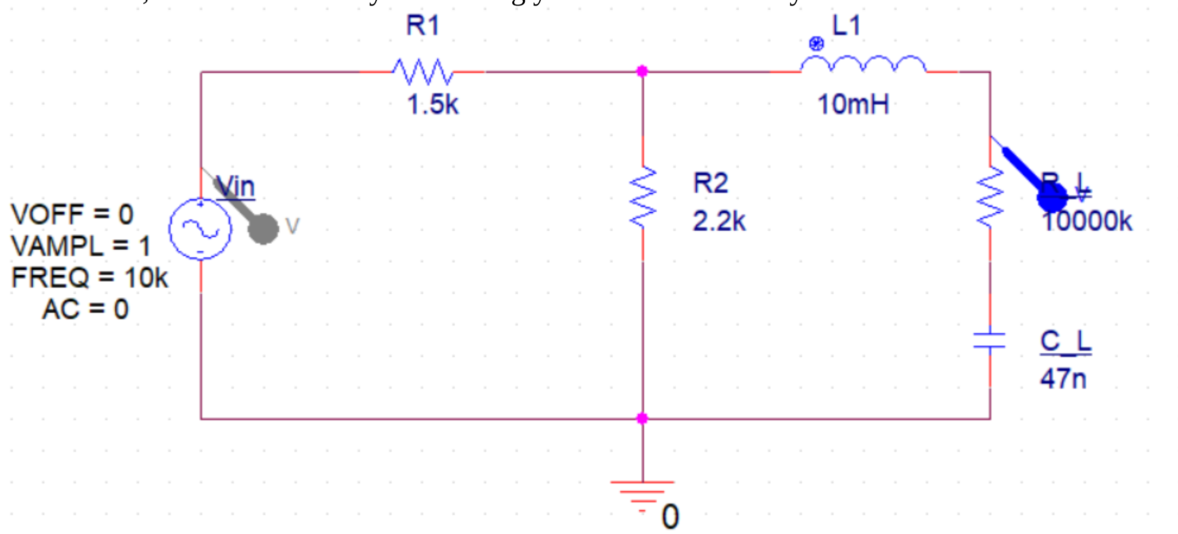


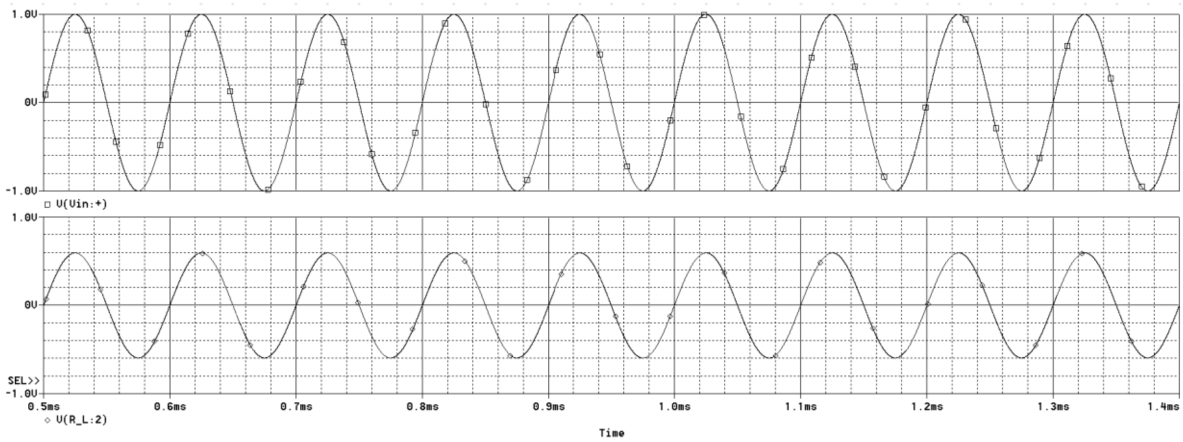


$V_L\angle\phi$ (Simulated) = 327.993m < -28.93

Δ t = -8.0371u

4. To measure Thevenin's voltage, instead of removing the load impedance replace the load with high value impedance (e.g., 10,000 kΩ keeping C_L value same), measure and record voltage across the load, Δt and its phase in given space below. Discard first 5 cycles then take measurement; this can be done by start saving your data after the 5 cycles.

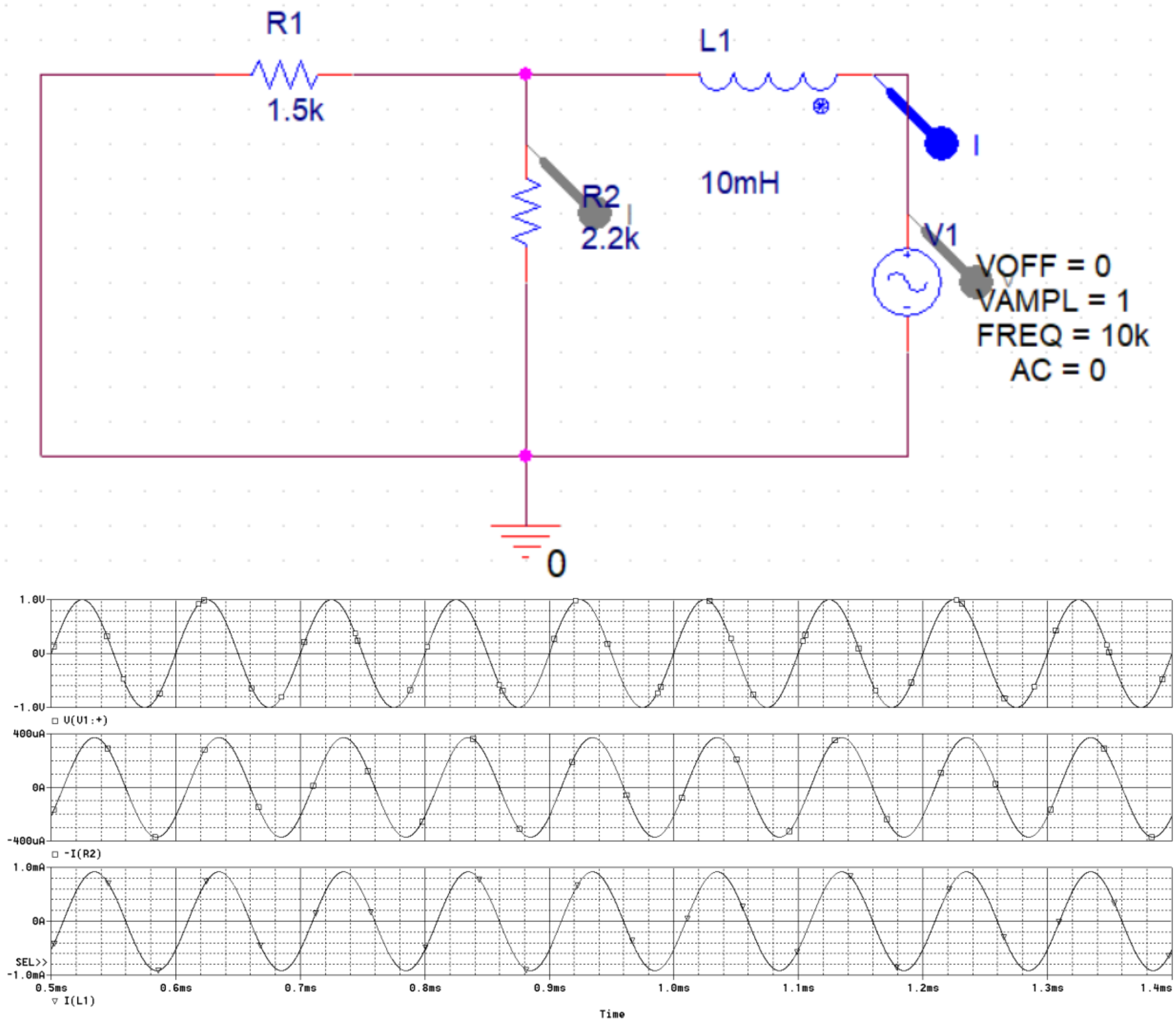




$V_{TH}\angle\phi$ (Simulated) = 594.542m<0.5

Δt = 140.055n

5. To find the Thevenin's resistance, replace the voltage source with a short wire, Replace the load Impedance (resistor and the capacitor) with voltage source providing 1V AC. Measure and record total current using marker at ***entering node- refer figure below*** and Δt (w.r.t V_{IN}). Discard first 5 cycles then take measurements.



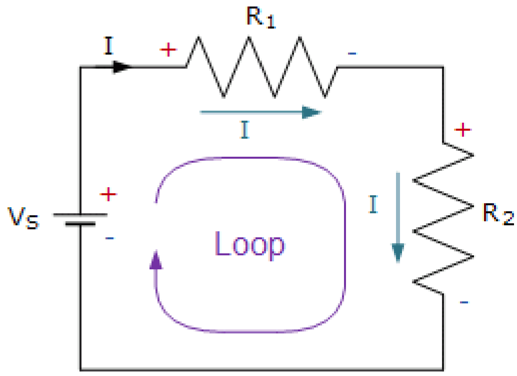


Figure 8.1a: Current is entering from negative terminal of the source, place your marker at -ve terminal

$I\angle\phi$ = 916.600u < -32.53
Δt = - 9.036u

6. Calculate the $Z_{TH}\angle\phi$ (Simulated) by using the formula ohms law.
   V = 1<0
   I = 916.600u < -32.53

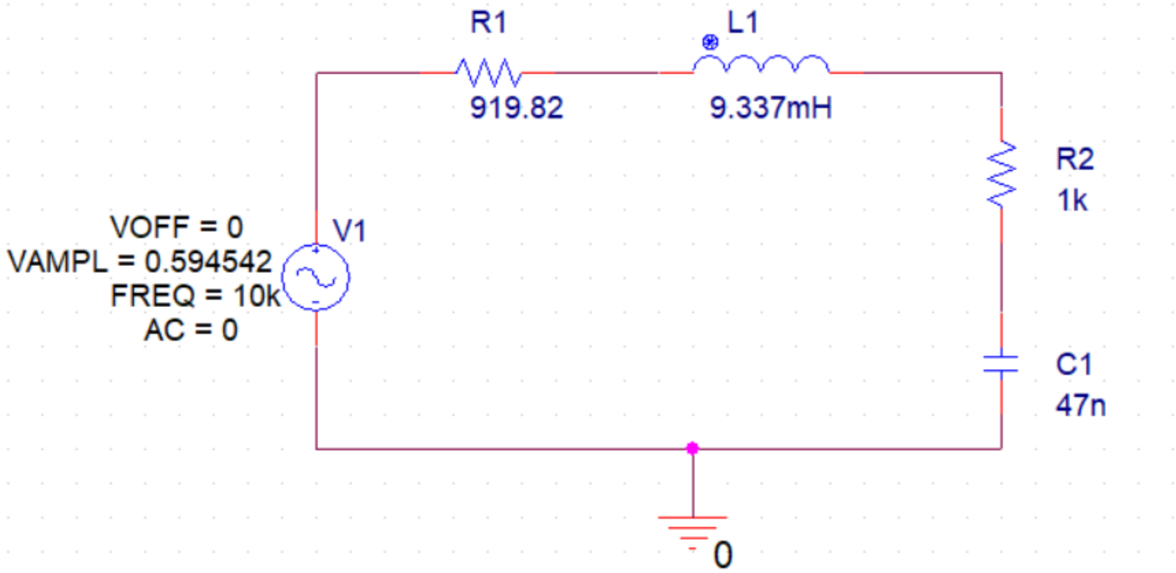
$Z_{TH}\angle\phi$ (Simulated) = 1090. 99 < 32.53
Z_{TH} (Simulated) = 919.82 + j586.67 <u>(rectangular form)</u>

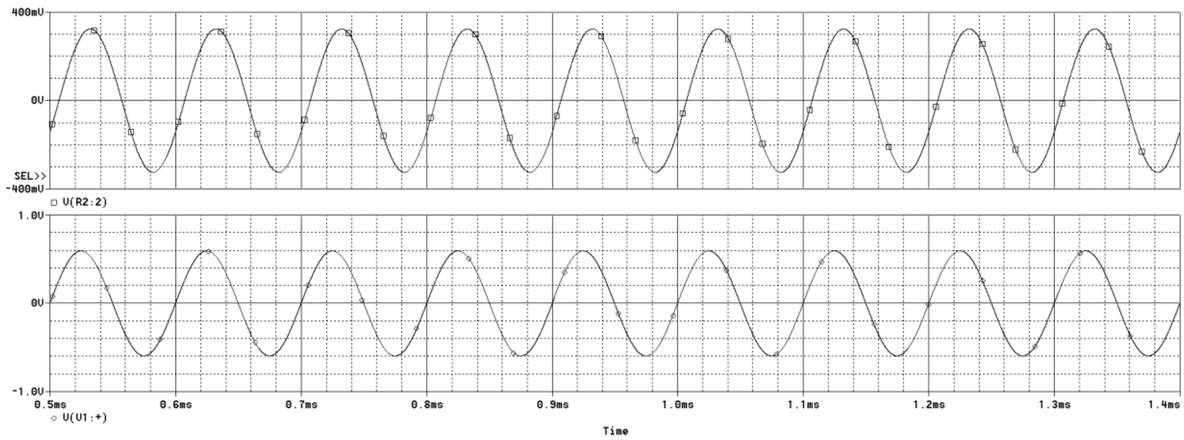
7. The sign of imaginary part of Z_{TH} will decide the nature of resultant component if the sign is positive then we calculate the value of inductor using formula $X_L = 2\pi f L$.

   586.67 = 2(3.142)(10000) L
   L = 9.34 mH

   L = 9.34 mH

8. Simulate the Thevenin's equivalent circuit using V_{TH} and Z_{TH} from previous steps, connect the load impedance (R_L=1kΩ and C_L=47nF). Measure the load voltage (V_L) record in given space.





$V_L\angle\phi$ (Simulated) = 324.265m < -26.5
Δt = -7.3634u

9. Attach simulation model Thevenin's equivalent circuit. Also, compare theoretical and simulated circuit values and determine the deviation between the original and Thevenin's equivalent circuit.

Table 8.1

	Calculate	Simulated	% Deviation
V_{TH}	0.5946<0	594.542m<0.5	0.009755%
Z_{TH}	1090.9<35.16 (891.9 + j628.3)	1090. 99 < 32.53 (919.82 + j586.67)	0.00825%
Vload	0.328 < -27.4	324.265m < -26.5	1.14%

There is barely any deviation proving my successful work during prelab and in simulations.

8.9 Norton's Theorem

This theorem states that any linear circuit containing several energy sources and resistances can be replaced by a single constant current generator in parallel with a single resistor. This is also the same as that of the Thevenin's theorem, in which we find Thevenin's equivalent voltage and resistance values, but here current equivalent values are determined.

Task 3: To simulate the Norton's current and impedance of the given AC Circuit.

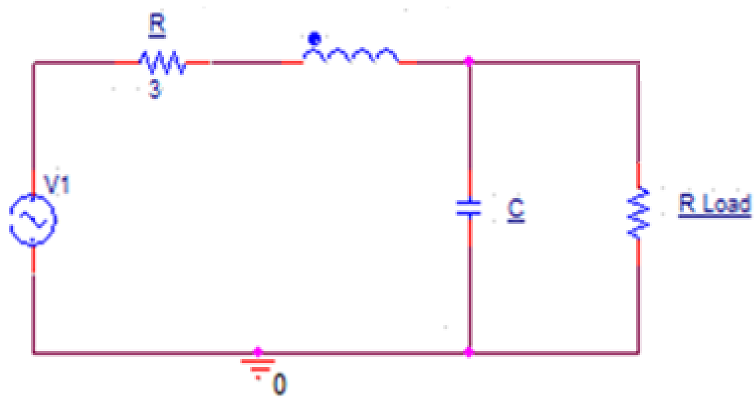
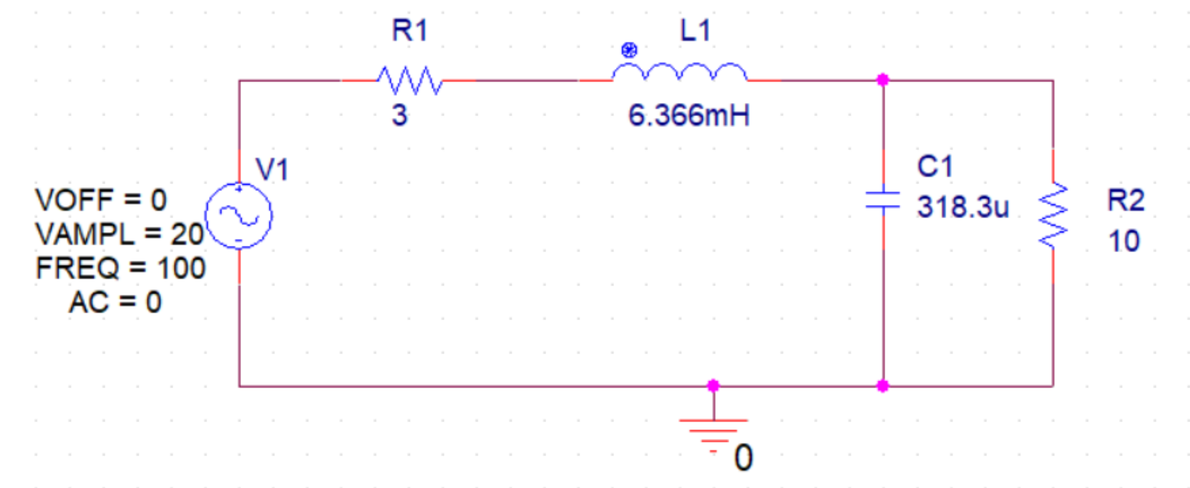


Figure 8.2: Schematic for AC circuit

Note: Attach snippet of every circuit and graph, all the graphs and circuits must be labelled properly.

1. For the circuit given in Figure 2, simulate Norton's Equivalent in PSpice.
   V1 = 20∠0,
   R = 3Ω,
   L = 6.366mH,
   C = 318.3uF,
   Frequency = 100 Hz
   5 < R_{LOAD} > 25 (take any value in the given range).
   I choose Rload = 10 ohms.

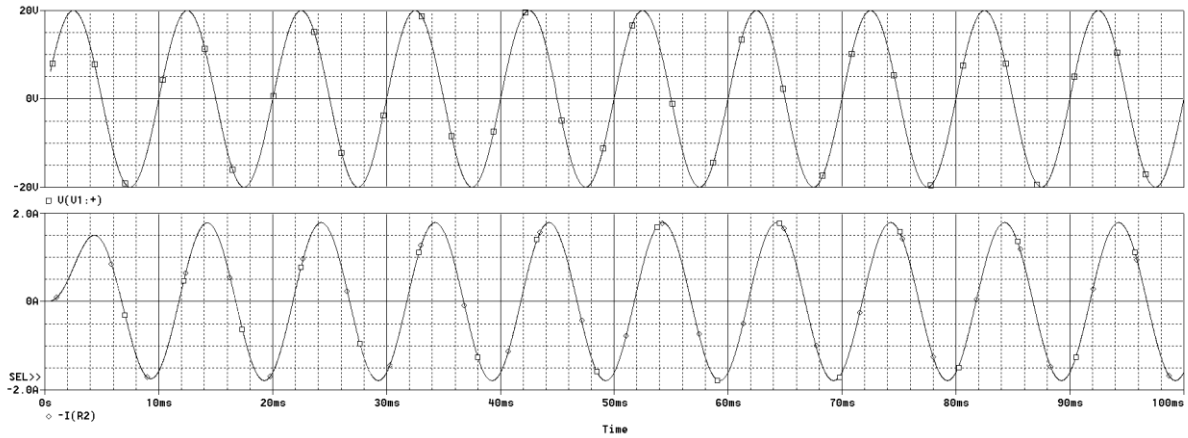


2. Verify Norton current I_N, Z_N and I_{LOAD} with the values calculated in pre lab exercise. For simulation of Z_N, remove the voltage source and add a test source in place of R_{LOAD} and find the circuit current, then find Z_N using the current and test source's voltage value.

For Iload:

Iload = 1.7889 < -63.342

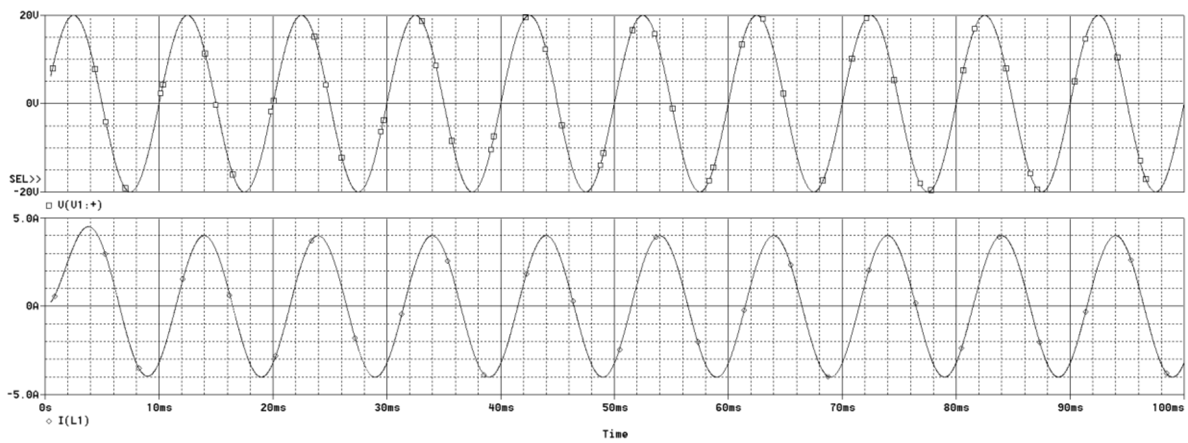
Delta t = -1.7595m



For In:

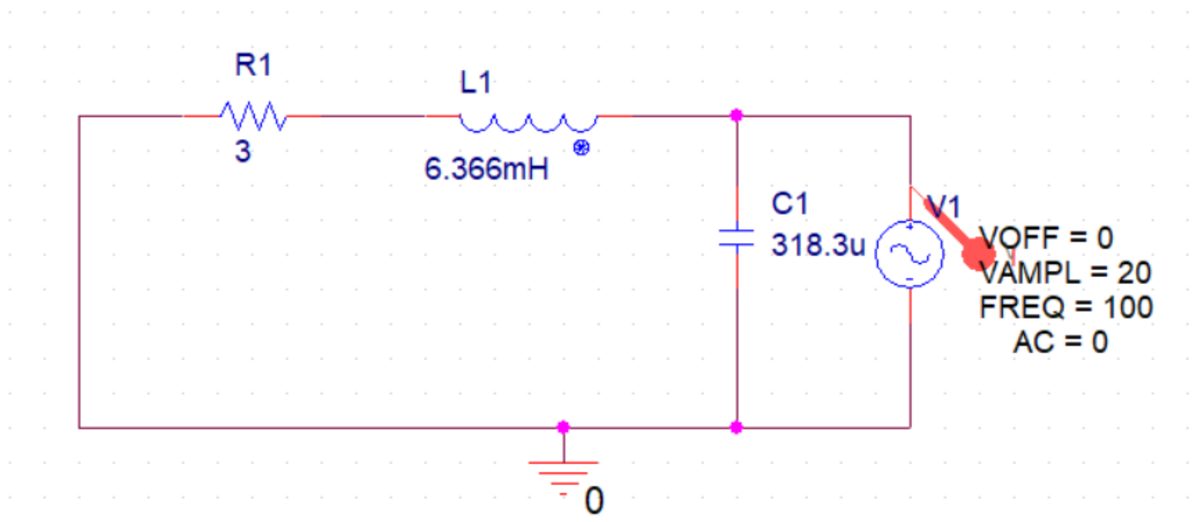
In = 4.0001 < -53.23

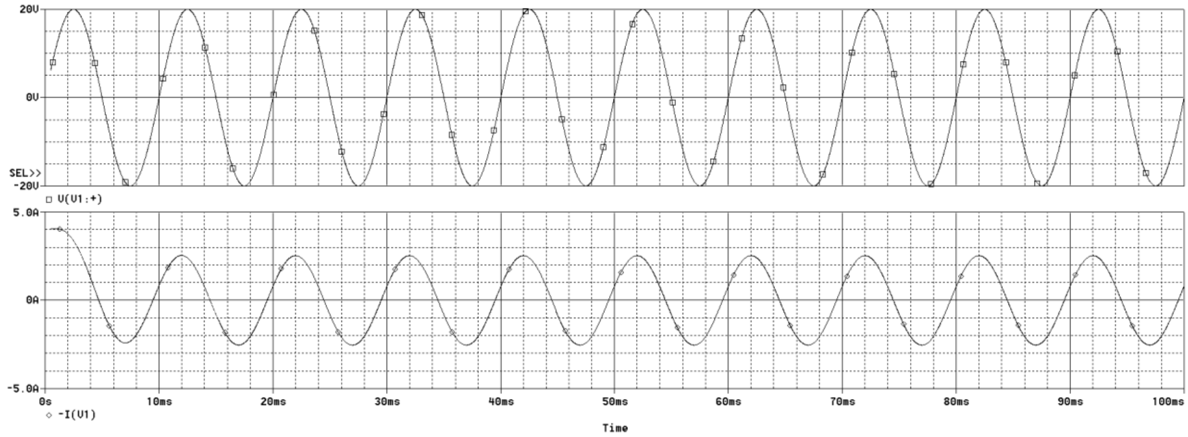
Delta t = -1.4785m



For Zn:

It = 2.5299 < 18.34
Delta t = 509.537u
From here Zn comes out to be = V/It
= (20 < 0) / (2.5299 < 18.34)
= 7.905 < -18.34





3. Attach the simulated circuit with clear and labelled graphs to support your answer. Attach circuit and simulations for all the major steps involved. *(Circuit simulation 5 points + results and plots 10 points)*

HINT: To set the Phase of AC source double click the source and enter the desired phase in the Phase column (just as we did to set initial condition in past labs). PSpice accepts phase values in degree.

Table 8.2

	Calculated	**Simulated**	**% Deviation**
I_N	4<-53.123 (2.4008 - j3.2)	4.0001 < -53.23	0.0025%
Z_N	7.905 < -18.43 (7.4997 - j2.4997)	*7.905 < -18.34*	0%

ILOAD	1.789 < -63.43 (0.8 - j1.5997)	1.7889 < -63.342	0%

Not much deviation going on in simulations and theory proving my successful math applied in prelab and in lab

8.4 Maximum Power Transfer Theorem

In the DC circuits, maximum power transfer is achieved by setting the load resistance equal to the source's internal resistance. This is not true for the AC case. Instead, the load should be set to complex conjugate of the source impedance. By using the complex conjugate, the load and source reactive components will cancel out leaving a purely resistive circuit similar to the DC case. When calculating the true load power, care must be taken to remember that the load voltage appears across a complex load impedance. Only the real portion of this voltage appears across the resistive component, and only the resistive component dissipates power.

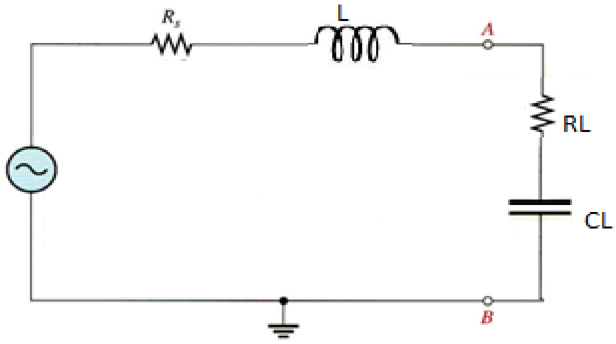


Figure 8.3: Schematic for an AC RLC circuit

Consider the circuit given in Figure 8.3, to transfer maximum power to the load terminals AB, the load connected here should be complex conjugate of the network impedance. The total impedance seen from the terminal AB is equal to the Thevenin Impedance.

Therefore, Z_{LOAD}=$Z_{TH}*$

If, $Z_{TH} = R + jX,$

then, $Z_{TH}* = R - jX$

The above equations implies that if the Thevenin impedance consists of inductive load (positive reactance) then, the load impedance should be capacitive (negative reactance) and vice versa. In the following steps, first you will find the Thevenin Impedance of the network as seen from the terminals AB. Then, you will connect different loads by varying the resistance and reactance to verify that the power transferred to the resistive component of the load is maximum when the load impedance is complex conjugate of the Thevenin Impedance.

Task 4: To verify the Maximum Power Transfer Theorem by varying load resistance and capacitance of the given AC Circuit.

Note: Attach snippet of every circuit and graph, all the graphs and circuits must be labelled properly.

1. Calculating the Thevenin Impedance**:** Consider the circuit of Figure 8.3. with R_s = 330Ω and L = 10mH but leaving off the load components. Replace the Input AC source with a short wire and calculate the total impedance seen from AB terminals. Let, the system frequency be 5 kHz. Express it in rectangular form.

   Z_{TH} (Calculated) = 330 + j314.16

2. Determine the load impedance at which maximum power transfer is achieved according to the theorem.

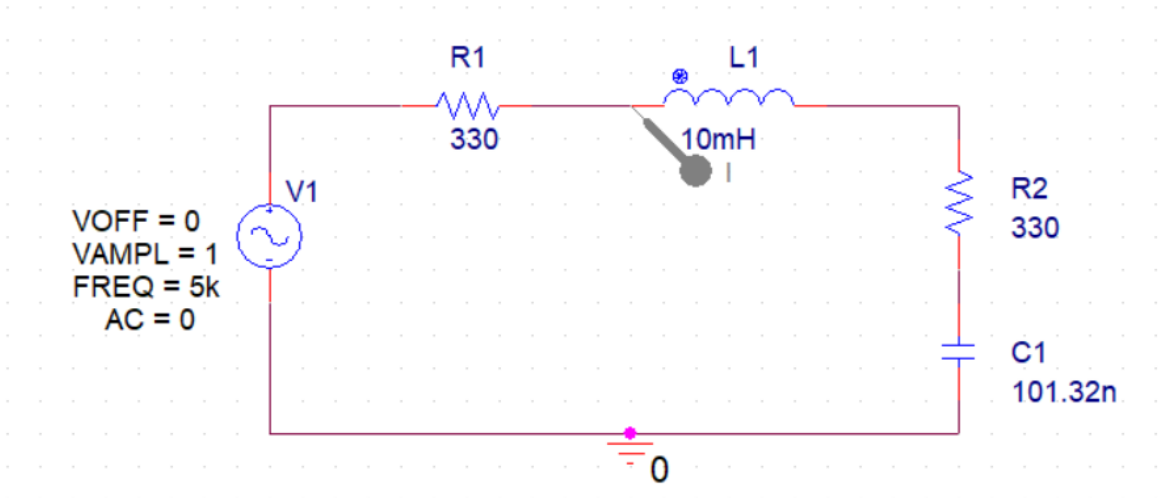
   Z_{LOAD} (Required) = 330 – j314.16

3. From Z_{LOAD} (Required), find the load resistance and capacitance; R_L and C_L.
   314.16 = 1/ 2(3.142)(5k) C
   C= 1/ 2(3.142) (5k) (314.16)
   C = 101.32nF

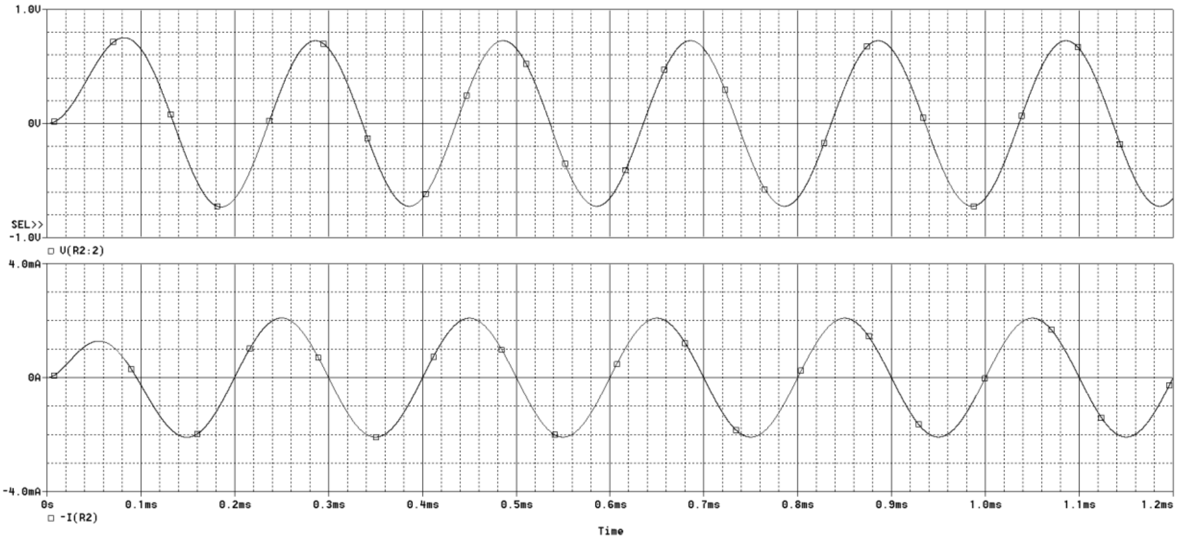
   R_L= 330

   C_L= 101.32nF

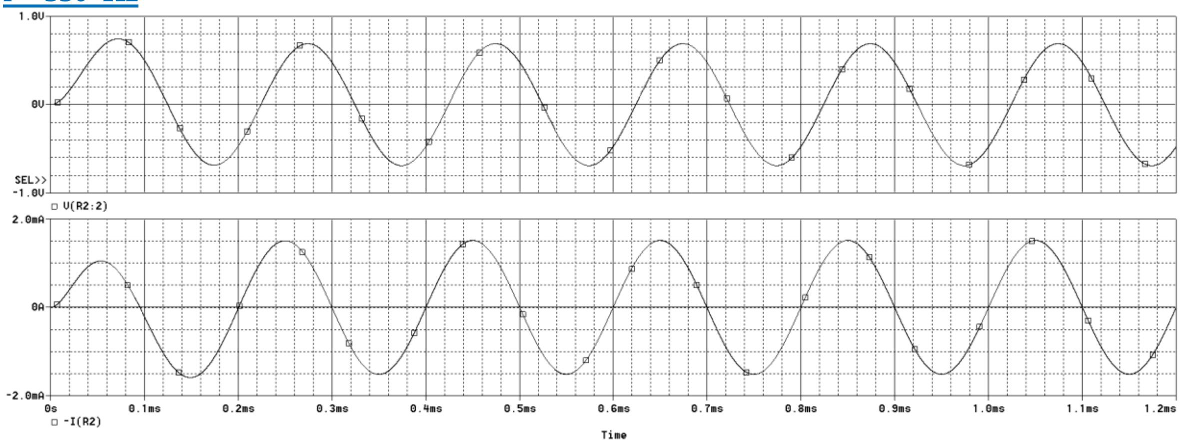
4. Simulate the circuit in Figure 8.3 on PSpice using the calculated values of load, take $V_S = 1V$ and $f = 5kHz$.



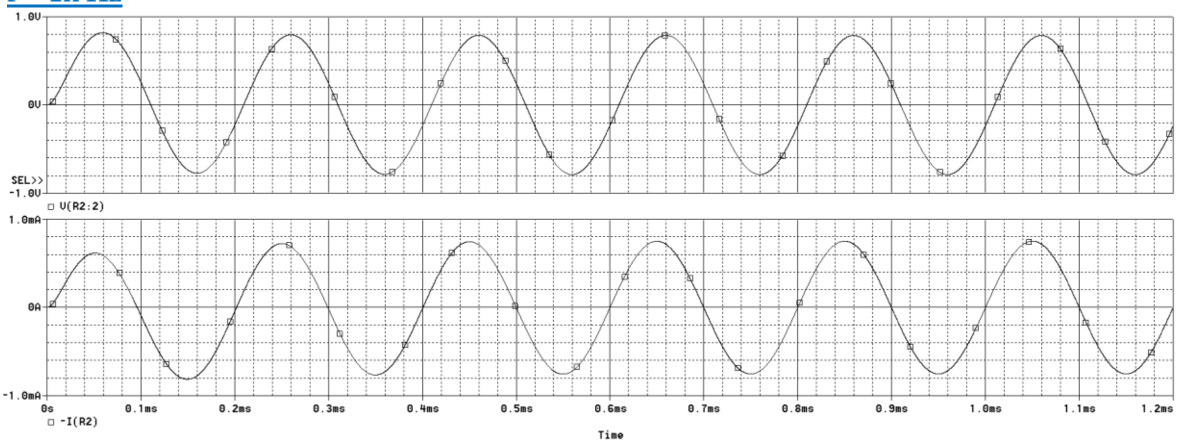
5. Now run the simulation for all the values of load resistance $R_L$listed in Table 8.3. keeping the load capacitance fixed. Record the peak value of voltage across load, peak value of current and their Δt (between I_p & V_p). Also, calculate the power across the load (P_{RL}) using I_{rms} & V_{rms} values.
   f = 150 Hz



- **f = 330 Hz**



- **f = 1k Hz**



-f = 3k Hz:

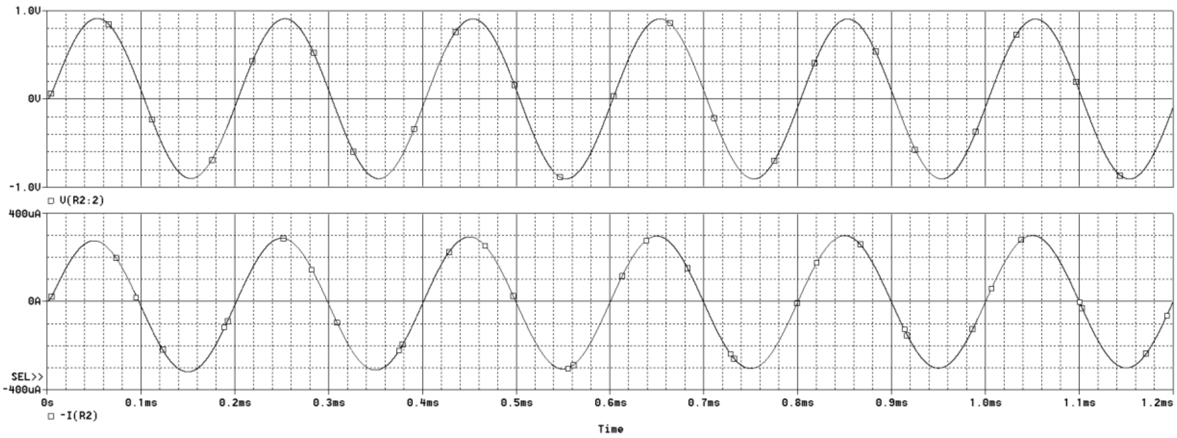
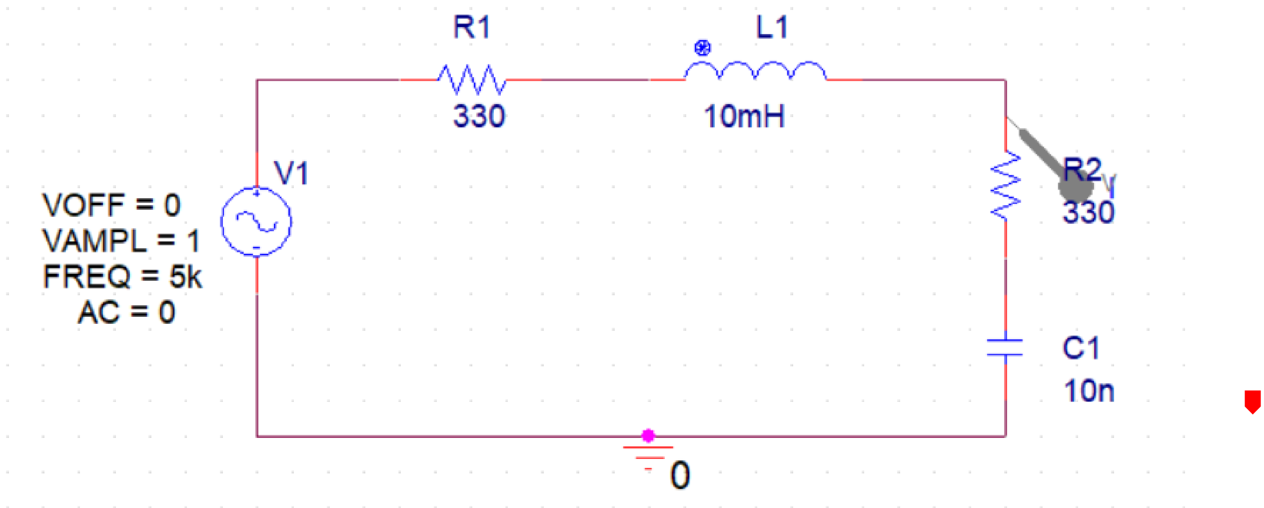


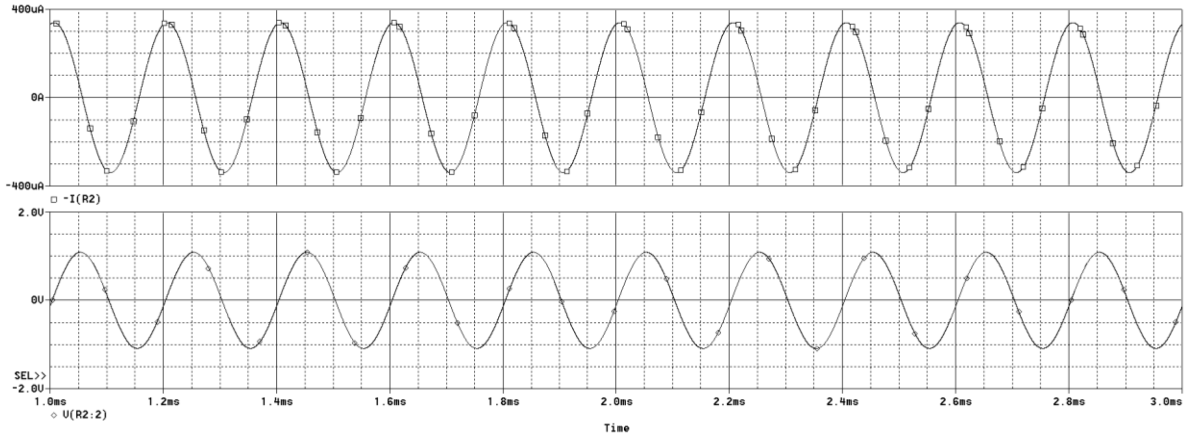
Table 8.3

$R_L(\Omega)$	I_p	I_{rms}	V_p	V_{rms}	Δt	θ	$P_{RL} = IV\cos(\theta)$
150	**2.0832m**	**1.473m**	**725.282m**	**512.85m**	**36.166u**	**65.099**	**318.085u**
330	**1.5151m**	**1.071m**	**690.348m**	**488.15m**	**23.876u**	**42.977**	**382.621u**
1k	**751.783u**	**531.59u**	**788.123m**	**557.29m**	**10.036u**	**18.065**	**281.628u**
3k	**299.052u**	**211.46u**	**906.198m**	**640.78m**	**-2.4656n**	**-0.0044**	**135.5u**

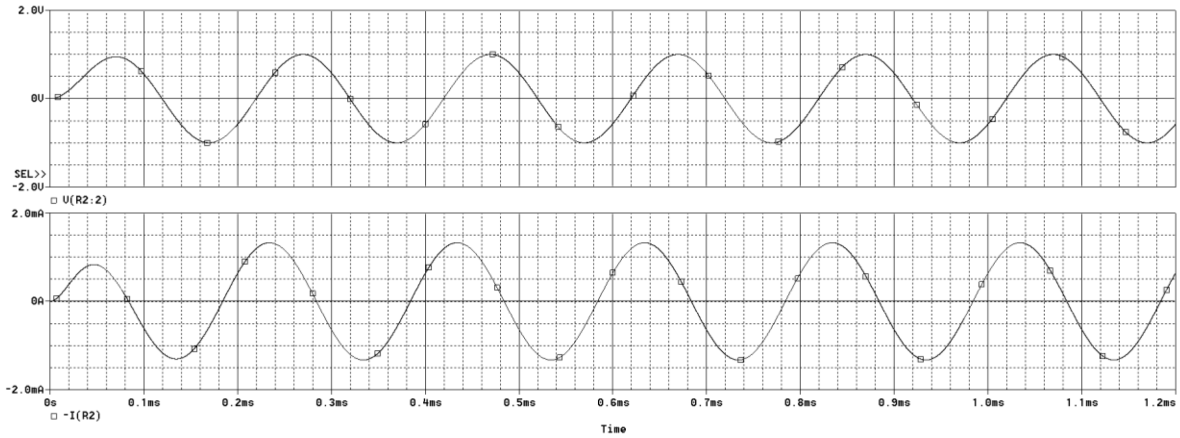
6. Now keeping load resistance fixed (as calculated in Table 8.3 for maximum transfer condition), run the simulation for all the values of C_L listed in Table 8.4.
   Schematic:



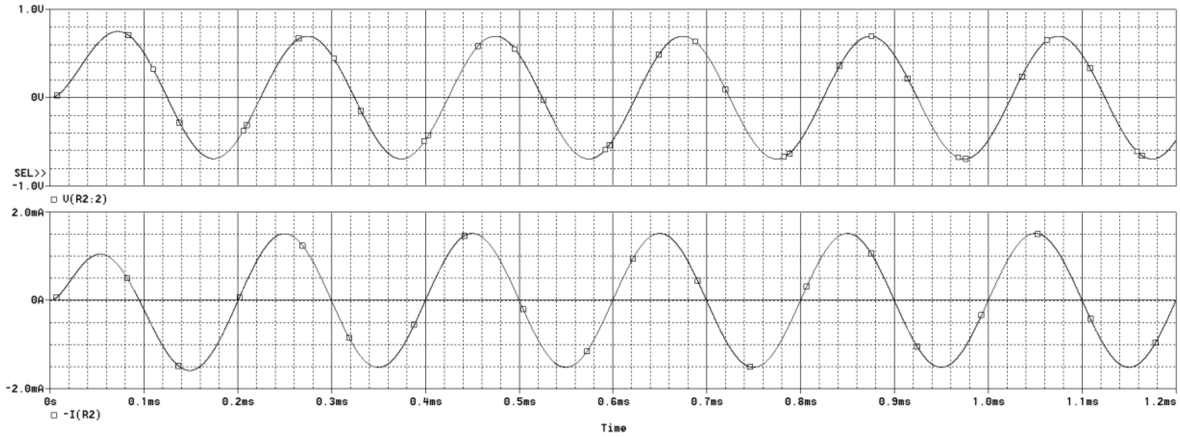
C = 10nF:



C = 47nF:



C = 100nF:



C = 1uF:

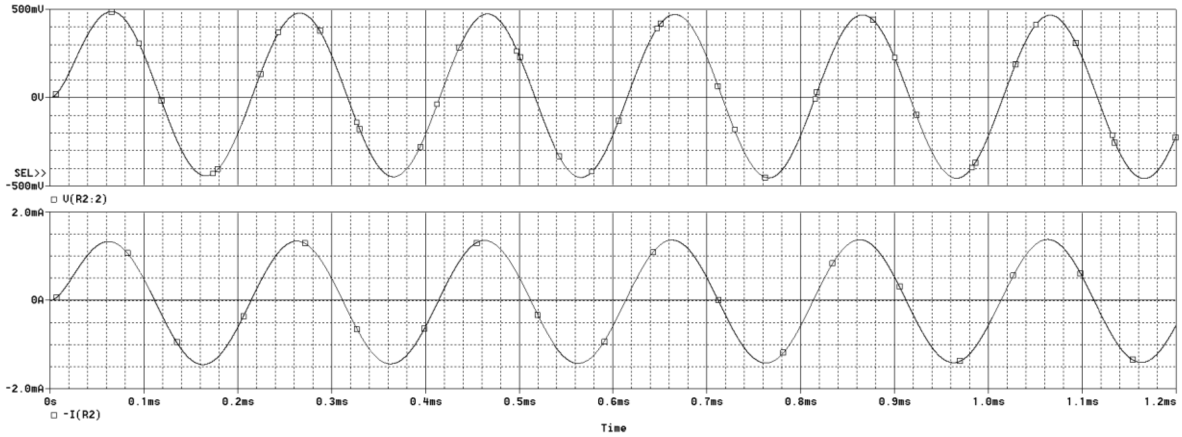


Table 8.4

C_L	I_p	I_{rms}	V_p	V_{rms}	Δt	θ	$P_{RL} = IVcos(\theta)$
10nF	339.688u	240.196u	1.0871	768.696m	45.335u	81.603	26.963u
47nF	1.3275m	936.68u	1.0001	707.177m	35.768u	64.3824	327.526u
100nF	1.5149m	1.0712m	691.567m	489.012m	26.906u	48.4308	347.572u
1uF	1.3780m	974.39u	466.552m	329.902m	3.1522u	5.674	319.879u

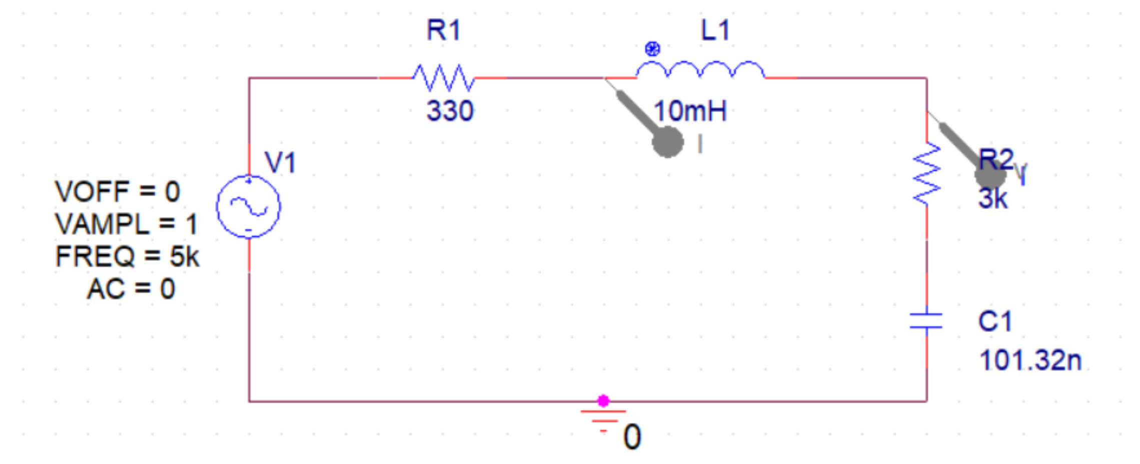
7. Analyze the data obtained in Table 8.4.

Analysis for Table 8.3:
The current in the circuit decreases as the load resistance increases. It's because less and less current is able to pass through the load resistor. Circuit reaches maximum power at RL = 330 ohms after which it begins to decrease. The peak power is obtained when the product of voltage and current is maximized and after this increasing the resistance only reduces more current leading to lesser power.

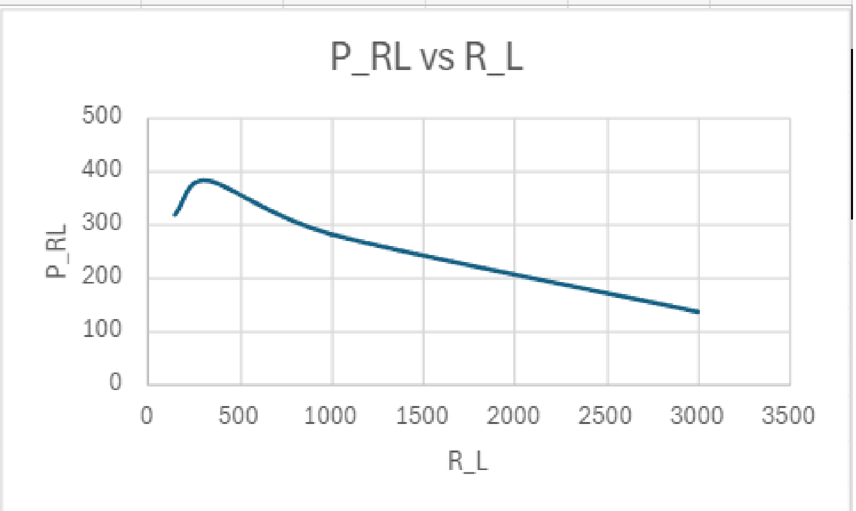
Analysis for Table 8.4:
With the increase in load capacitor, voltage decreases while the current increases until 100nF where it has reached maximum power and current starts decreasing again. A higher capacitance will store and release more energy per cycle. It allows or more current to pass through allowing more current to flow improving energy storage capability and increased power consumptions of the circuit.

8. Attach Simulation model and Plot graphs between P_{RL} Vs R_{Load} and P_{RL} Vs C_{Load} and provide your analysis on them.

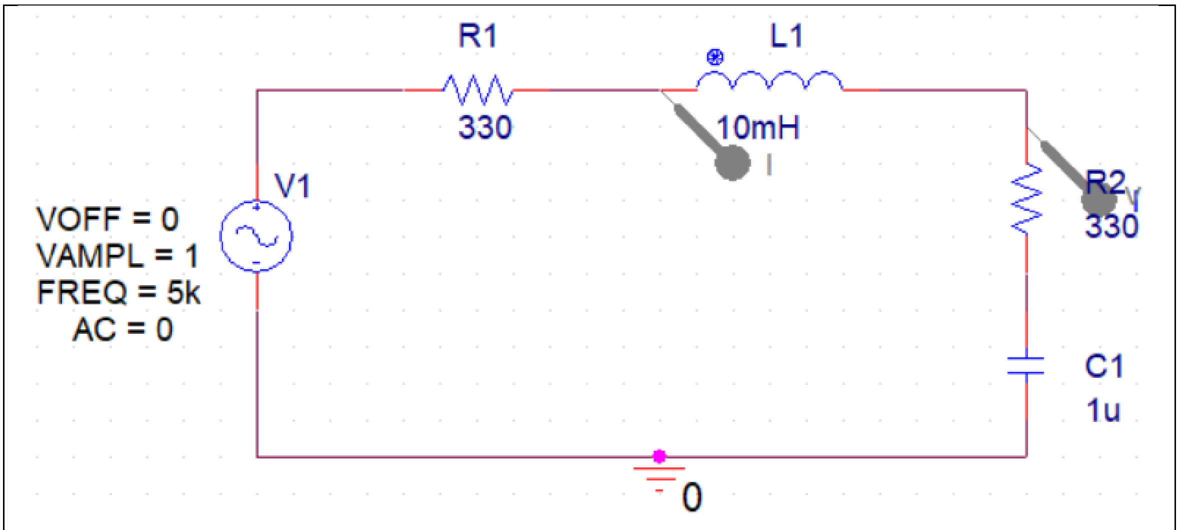
P_RL vs Rload



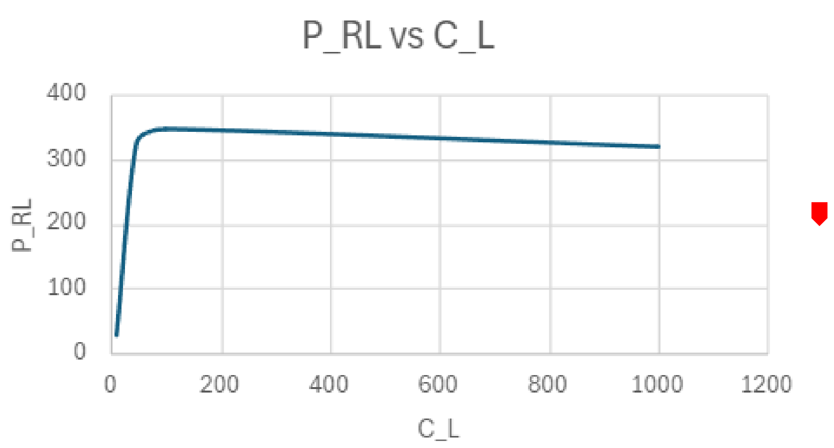
R_L	P_RL (uW)
150	318.085
330	382.621
1000	281.628
3000	135.5



P_RL vs C_load:



C_L (nF)	P_RL (uW)
10	26.963
47	327.526
100	347.572
1000	319.879



Assessment Rubric
Lab 08

Name:	**Student ID:**

Points Distribution

Task No.	LR 2 Simulation	LR5 Results/Plots	LR 6 Calculation	LR 10 Analysis
Pre-Lab – Task 1	-	-	/15	-
Task 2	/5	/15	-	-
Task 3	/5	/10	-	-
Task 4	/10	/20	-	/10
Total	/20	/45	/15	/10
CLO Mapped	CLO 3	CLO 3	CLO 3	CLO 3

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission (Pre-Lab+Report)	/(5+5)	CLO 4

CLO	Total Points	Points Obtained
3	90	
4	10	
Total	**100**	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR1	**Circuit Layout**	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR2	**Program/Code/ Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	**Troubleshooting**	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	**Data Collection**	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	**Calculations**	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR10	**Analysis**	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	**Design**	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.